VRANDESH BANDIKATTI

SENIOR AI ENGINEER | AGENTIC AI & MULTI-AGENT SYSTEMS | LLM / RAG | AI PLATFORM ENGINEERING | PYTHON

Enterprise Agentic AI | AI Co-workers | Tool-Calling | LangGraph / LangChain | Evaluation & Guardrails | FastAPI | Azure / AWS

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PROFILE

Senior AI Engineer and hands-on technology leader combining 16+ years of production software engineering with applied AI, LLM, RAG and agentic-system architecture. Experienced designing AI systems that reason across multi-step workflows, retrieve evidence, invoke tools, generate structured outputs, recover from failure and escalate material decisions through human-in-the-loop controls.

Designed and built multi-agent AI workflows for enterprise payroll assurance, regulatory document intelligence and governed natural-language data access. Hands-on experience across Python, FastAPI, LangGraph, LangChain, tool/function calling, MCP, structured outputs, hybrid retrieval, semantic chunking, re-ranking, evaluation, guardrails and LLM observability.

Strong focus on production-grade AI engineering: separating probabilistic model reasoning from deterministic business logic, designing bounded autonomous behaviour, evaluating quality and failure modes, maintaining source provenance and auditability, and implementing secure enterprise integration patterns. Brings deep software engineering foundations across distributed systems, APIs, event-driven architecture, cloud infrastructure, CI/CD, observability and security, enabling AI prototypes to be converted into reliable enterprise platforms.

Experienced leading engineers and working directly with business leaders, product teams and domain specialists to translate complex operational problems into scalable AI solutions.

AGENTIC AI, AI ENGINEERING & SOFTWARE PLATFORMS

Agentic AI & Multi-Agent Systems
LangGraph • LangChain • Multi-Agent Orchestration • Agentic Workflows • Tool Calling • Function Calling • MCP • Structured Outputs • Pydantic • JSON Schema • Prompt Engineering • Context Engineering • Self-Correction • Bounded Retries • Human-in-the-Loop • Controlled Autonomy

RAG & Knowledge Engineering
RAG • Hybrid Search • Vector Search • BM25 • Semantic Chunking • Embeddings • Vector Stores • Re-Ranking • Context Assembly • Source Attribution • Grounded Generation • Citation Preservation • Hallucination Detection • Retrieval Validation

AI Evaluation & Guardrails
Benchmarking • Regression Testing • Structured Output Validation • Failure-Mode Analysis • Guardrails • Eval Harness • Escalation Patterns • Human Approval Gates • Deterministic Validation

Machine Learning & NLP
Transformers • Embeddings • Semantic Similarity • Sentiment Analysis • Classification • Clustering • Recommendation Systems • Topic Modelling • Information Extraction • Supervised NLP • Sequence Modelling • Transformer Architecture • LSTM/CNN Fundamentals

AI Platforms & Models
Azure OpenAI • Amazon Bedrock • Anthropic Claude • LLM APIs • LangSmith • Model Orchestration • Model Evaluation • AI Observability

Python & AI Backend Engineering
Python • FastAPI • Django • Pydantic • Async Processing • REST APIs • GraphQL • Service-Layer Architecture • Validation • Structured Data Processing • Caching • Rate Limiting • Error Handling

Workflow & Distributed Systems
Azure Durable Functions • AWS Step Functions • Stateful Workflows • Parallel Execution • Event-Driven Architecture • Serverless • Microservices • Idempotency • Retry Policies • Failure Recovery • Long-Running Workflows • Graceful Degradation

Cloud & Platform Engineering
Azure • AWS • GCP • Azure Functions • AWS Lambda • Cloud Functions • Databricks • Microsoft Fabric • Azure SQL • Blob Storage • Firebase • Firestore • Docker • DEV/UAT/PROD

Data & Enterprise Integration
SQL • NoSQL • Power BI • Databricks • Microsoft Fabric • Xero • MYOB • Workday Adaptive Planning • API Integration • Batch Processing • Event-Driven Processing • Data Pipelines

Security & Governed AI
OAuth • OIDC • Entra ID • RBAC • Service Principals • Secrets Management • Least Privilege • PII Controls • Permission-Aware Processing • Human Approval • Audit Trails • Governed AI

Full Stack Engineering
TypeScript • JavaScript • Node.js • Next.js • React • Vue.js • REST • GraphQL • Enterprise Web Applications • CI/CD • Git • Docker • Automated Deployment • Environment Promotion • MLOps • Grafana • LangSmith • Tracing • Logging • Monitoring • Alerting • Incident Support

PROFESSIONAL EXPERIENCE

Senior Full Stack Developer - Data & AI Platform Architecture — Findex Australia

Melbourne · March 2022 – Present

- Architected and delivered production backend services and workflow-driven enterprise platforms using Python, Django, FastAPI, GraphQL, REST APIs and cloud-native services.
- Designed distributed processing workflows using Azure Durable Functions with parallel execution, retry policies, state tracking and idempotent activities for high-volume regulatory document processing.
- Built Django and FastAPI backend services with authentication, authorization, caching, rate limiting, structured query execution, bounded failure recovery, tracing and production alerting.
- Designed Python processing pipelines for complex financial, payroll and regulatory workloads, combining structured extraction, deterministic business rules, validation and exception processing.
- Established API and integration patterns across Power BI, Databricks, Microsoft Fabric, SharePoint, Workday Adaptive Planning, Xero, MYOB and internal enterprise services.
- Designed scheduled and automated data-processing workflows for financial modelling, benchmarking, student analytics and board-report generation.
- Defined engineering standards across API integration, service boundaries, CI/CD, security, observability, environment promotion and production support.
- Led cross-functional engineering teams while remaining hands-on with Python, Django, FastAPI, GraphQL, cloud integration and backend architecture.
- Stepped in as interim DevOps lead during a critical production incident, stabilising backend services, improving CI/CD reliability, adding Grafana observability and reducing manual operational overhead.

Senior Software Engineer, Product Innovation, Backend Cloud Engineering— Quest Payment Systems

Melbourne · October 2017 – March 2022

- Led backend and cloud engineering from concept through production for Airpay App, a regulated POS App and Donation Point Go, a regulated contactless donation platform.
- Architected and maintained 12+ production microservices and integration services using Node.js, AWS Lambda and AWS Step Functions.
- Designed event-driven, auto-scaling serverless architectures across AWS and GCP to handle unpredictable payment and transaction workloads.
- Built third-party integrations with Xero, MYOB and payment ecosystem services using resilient API integration patterns.
- Developed backend APIs supporting web, mobile, POS and kiosk applications.
- Implemented Google Firebase Functions, Firestore/real-time data services and NoSQL persistence for Airpay and Donation Point platforms.
- Led a team of three while remaining hands-on with backend development, cloud architecture, monitoring, logging and production support.

Senior Full Stack Developer – ESRI Australia

Melbourne · August 2017 – September 2017

- Developed enterprise geospatial and ArcGIS-based applications supporting location-based business solutions.
- Built AngularJS front-end modules for interactive enterprise applications.
- Integrated geospatial and enterprise services using SOAP/WSDL APIs.
- Developed Node.js / Express middleware to support application and service integration.
- Supported AWS-hosted infrastructure using EC2, Elastic Beanstalk, SQS and SNS.
- Implemented asynchronous processing and messaging workflows for distributed application components.
- Contributed across full-stack development, cloud deployment, systems integration and production support.

Technology Lead / Full Stack Developer — Apptarix Mobility

Bangalore · June 2013 – July 2017

- Filed a provisional patent application 201641021557 — a recommendation engine combining commercial and learnt rules, with a method to measure which of the two is earning its place.
- Built an early NLP sentiment engine using cosine similarity in R and Python; integrated clustering and sentiment analytics into OTT, analytics and EdTech product lines. Managed a team of 5. Integrated R and Python analytics for clustering and sentiment analysis, converting raw user data into product intelligence and customer insight.
- Designed scalable data ingestion, scraping and processing services that expanded the product data ecosystem and supported analytics-driven features.
- Managed a team of 5 in a startup environment using Agile and Scrum; led architecture across analytics dashboards, OTT platforms and client-facing product lines including Teletango, PrepSuite and MystiKit.

SELECTED AI AND ENTERPRISE PLATFORMS

Payroll AI — Agentic Payroll Assurance Platform
Findex · Solution Architect & Hands-on Engineer
Designed a four-stage multi-agent workflow covering regulatory document ingestion, structured rule extraction, client-specific calculation generation, payroll validation and exception remediation

- Built custom Python retrieval and chunking pipelines to preserve legal clause references across complex regulatory documents.
- Implemented end-to-end traceability from payroll outcomes back to originating award or agreement clauses, improving explainability and auditability.
- Designed controlled change workflows to assess the impact of rule changes across employees, clients and pay runs before approval.
- Automated exception classification and rerun coordination, reducing manual developer intervention in payroll corrections.
- Introduced human approval gates for material payroll issues and high-impact changes.
- Shifted payroll assurance from periodic audit activity toward continuous exception detection and remediation.

PerformX — Governed Enterprise AI Data Agent
Findex · Solution Architect & Hands-on Backend Engineer
Designed and delivered a production AI service that converts natural-language questions into validated queries against governed enterprise data while enforcing identity, permissions and data-access controls.

- Built the FastAPI orchestration layer covering identity resolution, query generation, validation, execution and response handling.
- Implemented Entra ID authentication and permission enforcement before generated queries could access financial data.
- Designed bounded self-recovery where failed generated queries use runtime errors to attempt correction, revalidation and safe retry.
- Added termination controls so unsuccessful recovery attempts fail safely rather than looping indefinitely.
- Ensured authoritative business values always came from the governed Power BI semantic model rather than being generated by the language model.
- Implemented rate limiting and metadata caching to improve performance and reduce unnecessary downstream platform calls.
- Added production tracing, monitoring, structured error handling and alerting. Exposed the service through both a React web application and Microsoft Teams.

WealthAI — AI Document Intelligence & Workflow Platform
Findex · Solution Designer & Full Stack Engineer
Designed a serverless platform for processing high volumes of regulatory documents using AI-assisted extraction, stateful orchestration and resilient workflow execution.

- Architected end-to-end processing using Azure Durable Functions with parallel execution, dependency management, retries and persistent workflow state.
- Built idempotent document-processing activities so failed workflows could be safely retried without duplicating downstream operations.
- Implemented structured schema validation to ensure extracted information met application contracts before further processing.
- Separated probabilistic AI extraction from deterministic business-rule execution to keep regulatory outcomes repeatable and explainable.
- Integrated workflow outputs with JSON, Excel and enterprise database services.
- Implemented workflow state tracking to support monitoring, troubleshooting and controlled reruns.

KRM Remuneration Platform — Enterprise Remuneration & Analytics Platform
Findex · Platform Architect & Delivery Lead
Architected the application, data and security foundations for an enterprise platform used to model and manage sensitive remuneration, incentive and workforce information.

- Designed processing pipelines supporting short-term incentives, referrals, forecasting and audit workloads across enterprise financial datasets.
- Implemented Entra ID SSO, RBAC and row-level security across application services and Power BI Embedded.
- Designed backend authorization controls so users could only access permitted financial and workforce information.
- Established governed bronze, silver and gold data layers using Databricks, Microsoft Fabric and OneLake.
- Built the remuneration calculation engine covering performance thresholds, payaway curves, gateways, caps, floors and employee segmentation.

- Replaced spreadsheet-driven remuneration calculations with a deterministic and auditable processing engine.
- Supported production onboarding for 443 users and 129 people leaders under the platform's role and permission model.

EDUCATION, CERTIFICATIONS & IP

- Harvard Business School Online — Certificate in Business Analytics, 2023
- BE, Electronics & Communication — Visvesvaraya Technological University, First Class Honours, 2008
- Certifications — Azure for Data Engineering (Coursera); Machine Learning & Big Data; Generative AI; React with TypeScript. PMI Melbourne member. PMP, Anthropic Claude Developer, Snowflake and Databricks pathways in progress.